

GRIDLY: SEARCHING THE HISTORY OF FORMULA 1

A PRI PROJECT

INDEX

1) DATASET

2) PIPELINE

3) CONCEPTUAL MODEL

4) DATA CHARACTERIZATION

5) SEARCH SCENARIOS



1

DATASET

DESCRIPTION AND ORIGIN

- THE DATASET WAS CONSTRUCTED USING TWO COMPLEMENTARY SOURCES : JOLPICA F1 DATABASE AND ENGLISH WIKIPEDIA

Source	Format	Files	Rows	Columns	Size (MB)
Jolpica F1	CSV	11	141029	110	7.30
Wikipedia	JSON	2225	2225	7	77.37

DESCRIPTION AND ORIGIN

- IN OUR FINAL COLLECTION WE HAVE A TOTAL OF 2058 DOCUMENTS DIVIDED IN FOUR TYPES: RACES, DRIVERS, TEAMS, AND CIRCUITS
- EACH DOCUMENT COMBINES STRUCTURED FIELDS FROM JOLPICA AND TEXT FIELDS FROM WIKIPEDIA. IN EACH OF THEM WE CAN ALSO FIND THE URL TO THE WIKIPEDIA ARTICLE

	Races	Drivers	Teams	Circuits	Total
Jolpica records	1162	818	168	77	2225
Accepted	1116	719	147	76	2058
Rejected	46	99	21	1	167
Fields	34	27	24	25	59
Median words	827	690	1082	1238	–



2

PIPELINE

PIPELINE

Jolpica F1

Download and unpack database

Data Ingestion

Relational tables (CSV)

Entity Structuring

Entity records & wiki targets

Wikipedia Crawler

Pass raw wikitext for cleaning

Wikitext Parser

Strip templates & extract plain text

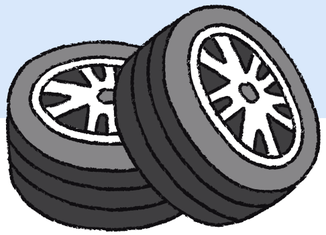
Schema Mapping & Filtering

Filter & map schema

Relational Linker

Build unified collection

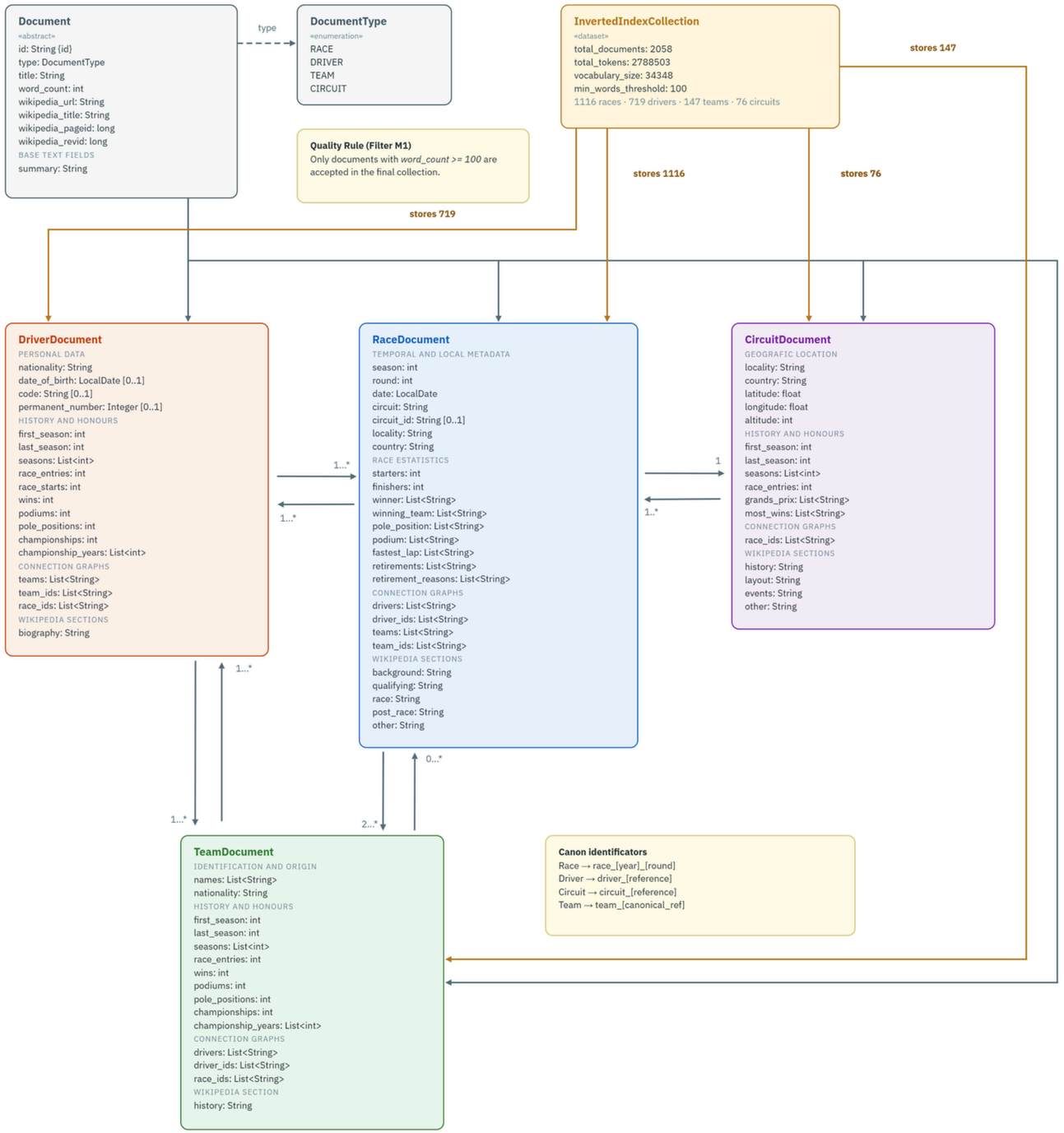
Collection Characterization





3

CONCEPTUAL MODEL


Document
«abstract»
id: String {id}
type: DocumentType
title: String
word_count: int
wikipedia_url: String
wikipedia_title: String
wikipedia_pageid: long
wikipedia_revid: long
BASE TEXT FIELDS
summary: String
type
DocumentType
«enumeration»
RACE
DRIVER
TEAM
CIRCUIT
InvertedIndexCollection
«dataset»
total_documents: 2058
total_tokens: 2788503
vocabulary_size: 34348
min_words_threshold: 100
1116 races · 719 drivers · 147 teams · 76 circuits
stores 147
Quality Rule (Filter M1)
Only documents with word_count >= 100 are accepted in the final collection.
stores 1116
stores 76
stores 719
DriverDocument
PERSONAL DATA
nationality: String
date_of_birth: LocalDate [0..1]
code: String [0..1]
permanent_number: Integer [0..1]
HISTORY AND HONOURS
first_season: int
last_season: int
seasons: List<int>
race_entries: int
race_starts: int
wins: int
podiums: int
pole_positions: int
championships: int
championship_years: List<int>
CONNECTION GRAPHS
teams: List<String>
team_ids: List<String>
race_ids: List<String>
WIKIPEDIA SECTIONS
biography: String
1...*
1...*
RaceDocument
TEMPORAL AND LOCAL METADATA
season: int
round: int
date: LocalDate
circuit: String
circuit_id: String [0..1]
locality: String
country: String
RACE ESTATISTICS
starters: int
finishers: int
winner: List<String>
winning_team: List<String>
pole_position: List<String>
podium: List<String>
fastest_lap: List<String>
retirements: List<String>
retirement_reasons: List<String>
CONNECTION GRAPHS
drivers: List<String>
driver_ids: List<String>
teams: List<String>
team_ids: List<String>
WIKIPEDIA SECTIONS
background: String
qualifying: String
race: String
post_race: String
other: String
1
1..*
CircuitDocument
GEOGRAFIC LOCATION
locality: String
country: String
latitude: float
longitude: float
altitude: int
HISTORY AND HONOURS
first_season: int
last_season: int
seasons: List<int>
race_entries: int
grands_prix: List<String>
most_wins: List<String>
CONNECTION GRAPHS
race_ids: List<String>
WIKIPEDIA SECTIONS
history: String
layout: String
events: String
other: String
1...*
1...*
0...*
2...*
TeamDocument
IDENTIFICATION AND ORIGIN
names: List<String>
nationality: String
HISTORY AND HONOURS
first_season: int
last_season: int
seasons: List<int>
race_entries: int
wins: int
podiums: int
pole_positions: int
championships: int
championship_years: List<int>
CONNECTION GRAPHS
drivers: List<String>
driver_ids: List<String>
race_ids: List<String>
WIKIPEDIA SECTION
history: String
Canon identificators
Race → race_[year]_[round]
Driver → driver_[reference]
Circuit → circuit_[reference]
Team → team_[canonical_ref]

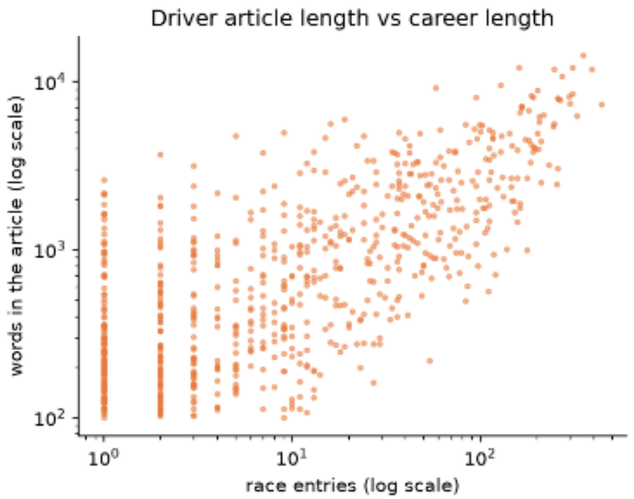


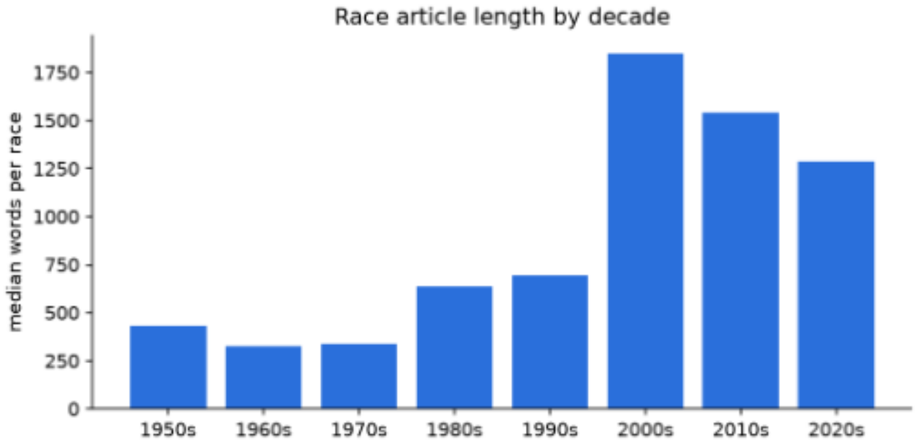
4

DATA CHARACTARIZATION

DATA CHARACTERIZATION

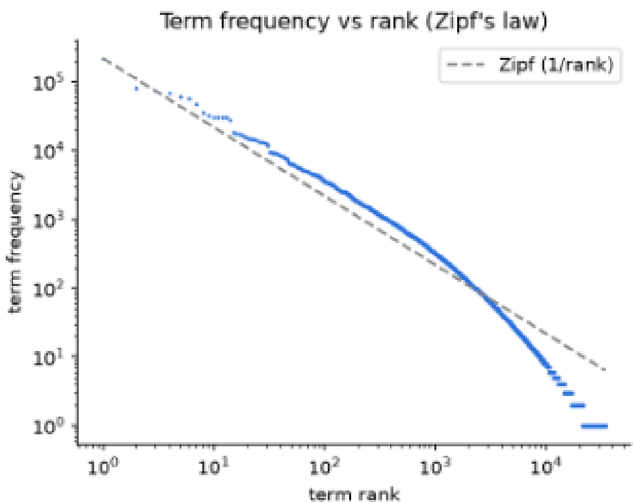
- THE LENGHT OF EACH DOCUMENT IS VERY UNEVEN - THE MEAN IS WELL ABOVE THE MEDIAN
- FOR THE DRIVERS THE LENGHT OF THE DOCUMENT GROWS WITH THE CAREER - SPEARMAN CORRELATION OF 0.69
- RACE PAGES ARE FAR LONGER IN RECENT DECADES: THE MEDIAN ROSE FROM ROUGHLY 330 WORDS IN THE 1960S TO 1,849 IN THE 2000S.

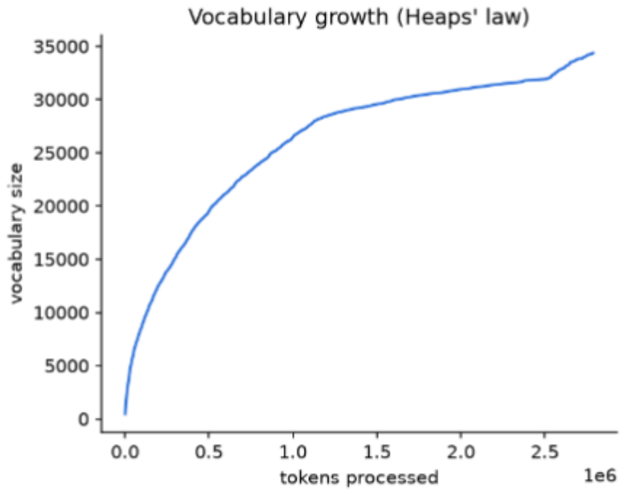




DATA CHARACTERIZATION

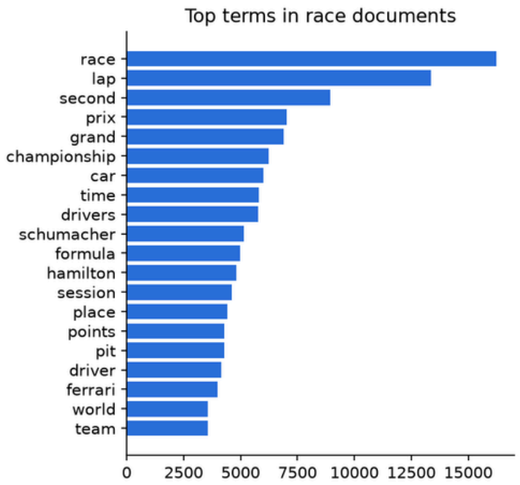
- OUR COLLECTION HAS 2.79 MILLION TOKENS AND 34348 TERMS (37% ONLY APPEAR ONCE)
- 45% OF OUR TOKENS ARE STOPWORDS
- THE TERM FREQUENCY FOLLOWS THE ZIPF'S LAW BETWEEN RANKS 10 AND ABOUT 2000
- THE VOCABULARY GROWS SUBLINEARLY - HEAPS' LAW

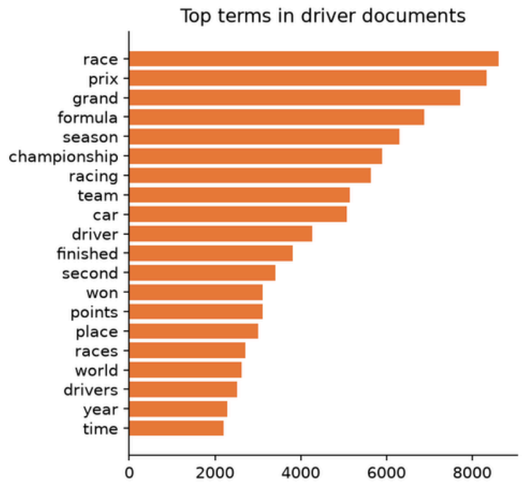


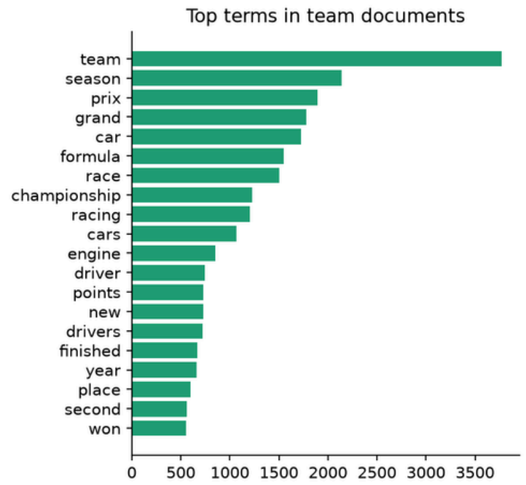


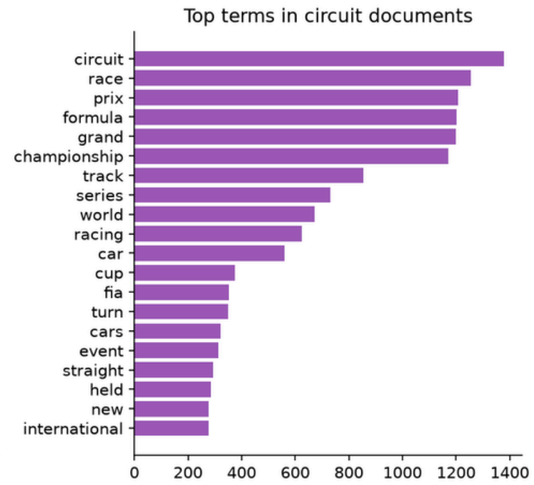
DATA CHARACTERIZATION

- TERMS LIKE : CHAMPIONSHIP, PRIX, RACE, ... APPEAR IN ABOUT 95% OF THE DOCUMENTS SO THEY WILL HAVE LITTLE WEIGHT IN THE RANKING
- THE TERMS ARE SPECIFIC TO EACH TYPE OF DOCUMENT



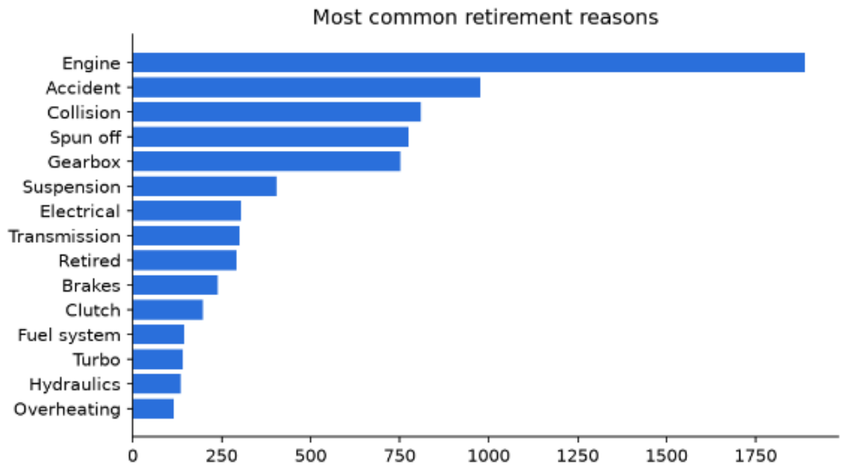


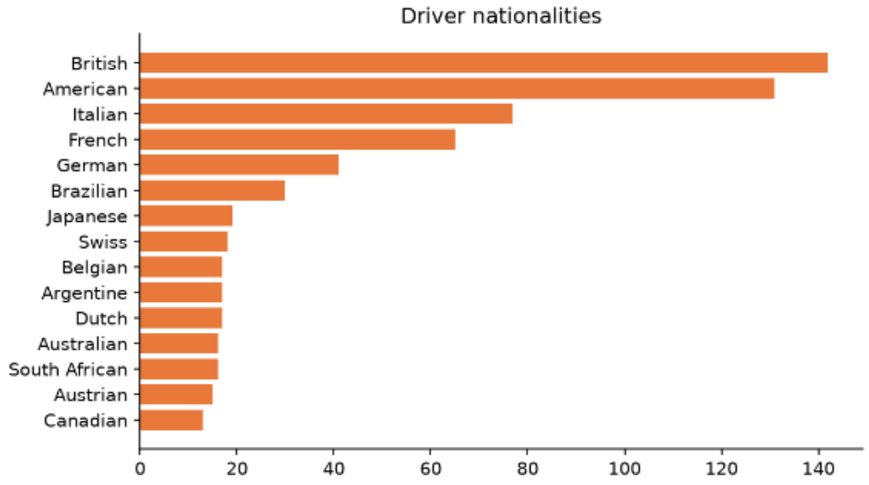




DATA CHARACTERIZATION

- MOST COMMON RETIREMENT REASON: ENGINE FAILURE
- MOST COMMON NATIONALITIES: BRITISH AND AMERICAN
- FOR EVERY RACE IN OUR DATA SET WE HAVE A SUMMARY, BUT ONLY HAVE A RACE REPORT FOR 85% OF THE RACES







5

SEARCH SCENARIOS

SEARCH SCENARIOS

- AS A USER I WANT TO BE ABLE TO SEARCH RACES WHERE A **SAFETY CAR** HAPPENED AND HAD AN IMPORTANT IMPACT ON THE **RACE RESULT**
- AS A USER I WANT TO BE ABLE TO LOOK UP IF TWO DRIVERS ALREADY **COLLIDED** WHILE BEING **TEAMMATES**
- AS A USER I WANT TO BE ABLE TO SEARCH WHICH RACE WEEKEND HAD A FATAL **ACCIDENT**
- AS A USER I WANT TO BE ABLE LOOK UP WHICH **TECHNICAL INNOVATIONS** WERE LATER ON **BANNED**

ANY QUESTIONS?



Gridly

F1 SEARCH SYSTEM